

HW 1

Problem 1

Let G be a group and $N \trianglelefteq G$

Let ψ be an homomorphism from G to G_1 s.t. $N \subseteq \ker(\psi)$

Define: $\bar{\psi} : G/N \rightarrow G_1$ s.t. $\bar{\psi}(gN) = \psi(g)$

Well Defined: Suppose $gN = hN \implies gh^{-1} \in N \implies gh^{-1} \in \ker(\psi)$

$\psi(gh^{-1}) = e \implies \psi(g) = \psi(h)$ since ψ is a homomorphism this implies $\bar{\psi}(gN) = \bar{\psi}(hN)$

Homomorphism: Let $gN, hN \in G/N$

$\bar{\psi}(gNhN) = \bar{\psi}(ghN) = \psi(gh) = \psi(g)\psi(h) = \bar{\psi}(gN)\bar{\psi}(hN)$

Uniqueness: Suppose there exists another homomorphism $\phi : G/N \rightarrow G_1$ such that $\phi = \psi \circ \varphi$

$$\phi(gN) = \psi(g) \implies \phi = \bar{\psi}$$

Problem 2

Let H be a subgroup of G .

- (Inverses) Let $a \in N_G(H)$
Consider: $a^{-1}Ha = a^{-1}aH = H$ thus $a^{-1} \in N_G(H)$
- (Closure) Let $a, b \in N_G(H)$
Consider: $abHb^{-1}a^{-1} = abb^{-1}Ha^{-1} = aHa^{-1} = H$ thus $ab \in N_G(H)$
- (Identity) Consider $eHe = H$ thus $e \in N_G(H)$

Let $h \in H$

Consider : hHh^{-1} since $h, h^{-1} \in H \implies hHh^{-1} = H$ thus $h \in N_G(H)$ thus $H \subseteq N_G(H)$

By definition since $\forall g \in N_G(H) gHg^{-1} = H$ thus $H \trianglelefteq N_G(H)$

Largest: Let K be a subgroup of G s.t. $H \trianglelefteq K$ thus $\forall k \in K$

$$kHk^{-1} = H$$

thus $k \in N_G(H)$ thus $K \subseteq N_G(H)$

Problem 3

- (i)

Proof.

- (Closure) Let $hk, jl \in HK$ s.t. $h, j \in H$ and $k, l \in K$
 $hkjl = h(kj)l$ since $H \subseteq N_G(K) \implies kj = jk'$ for $k' \in K$
 $h(jk')l = hjk'l \in HK$
- (Identity) Since H, K are subgroups of G , $e \in H \cap K \implies ee = e \in HK$
- (Inverses) Let $hk \in HK$ s.t. $h \in H$ and $k \in K$
 Consider $k^{-1}h^{-1}$ since $H \subseteq N_G(K) \implies k^{-1}h^{-1} = h^{-1}k'$ for $k' \in K$
 thus $k^{-1}h^{-1} = h^{-1}k' \in HK$
- Let $hk \in HK$ s.t. $h \in H$ and $k, k' \in K$
 $(hk)k'(hk)^{-1} = h(kk'k^{-1})h^{-1}$ thus $kk'k^{-1} = l \in K$ and since $h \in H \subseteq N_G(K)$
 $\implies hllh^{-1} \in K$ thus $(hk)k'(hk)^{-1} \in K$ thus $K \trianglelefteq HK$

□

- (ii)

Proof. Let $l \in H \cap K \implies l \in H$ and $l \in K$ and $h \in H$
 hlh^{-1} since $l \in K$ and $H \subset N_G(K) \implies hlh^{-1} \in K$
 since $h \in H$ and $l \in H \implies hlh^{-1} \in H$ thus $hlh^{-1} \in H \cap K$ thus $H \cap K \trianglelefteq H$

□

- (iii)

Proof. Let $\varphi : H \rightarrow HK/K$ s.t. $\varphi(h) = hK$
 Well-defined: Suppose $h_1 = h_2 \in H \implies \varphi(h_1) = h_1K = h_2K = \varphi(h_2)$
 Homomorphism: Let $h, j \in H$
 $\varphi(hj) = hjK = hKjK = \varphi(h)\varphi(j)$
 Kernel: Let $h \in \ker(\varphi) \trianglelefteq H \implies h \in H$
 $\varphi(h) = hK = K \implies h \in K \implies h \in H \cap K \implies \ker(\varphi) = H \cap K$
 Surjective: Let $hkK \in HK/K$, for $h \in H, k \in K$
 $\varphi(h) = hK = hkK$ thus $\text{im}(\varphi) = HK/K$
 Thus by the first isomorphism theorem $H/H \cap K \cong HK/K$

□

Problem 4

- (a) • (i. $\implies$ ii.)

Assume i.

Let $g \in G$ by i. we know $g = hk$ for $h \in H$ and $k \in K$ thus $g = hk \in HK$

Let $hk \in HK$ s.t. $h \in H$ and $k \in K$ since $H, K \subseteq G \implies hk \in G$ thus $G = HK$

Since $e \in H, K \implies e \in H \cap K \implies \{e\} \subseteq H \cap K$

Let $l \in H \cap K \implies l \in H$ and $l \in K$ and $g \in G$ by i. $g = hk$ for $h \in H$ and $k \in K$

$g = hxx^{-1}k$ thus $hx \in H$ and $x^{-1}k \in K$ thus by i. since we assume uniqueness for each $g \in G$ this implies $x = e$ thus $H \cap K = \{e\}$

- (ii. $\implies$ iii.)

Assume ii.

Define: $\varphi : H \times K \rightarrow G$ s.t. $\varphi(h, k) = hk$

Homomorphism: Let $(h, k), (j, l) \in H \times K$

$\varphi((h, k)(j, l)) = \varphi(hj, kl) = hjkl = \varphi(hj)\varphi(kl)$

Injective: Let $(h, k) \in \ker(\varphi)$

$\varphi(h, k) = hk = e \implies h = k^{-1} \implies h, k \in H \cap K$ by ii. $\implies h = k = e$

Thus $(h, k) = (e, e) \implies \ker(\varphi) = \{(e, e)\} \implies \varphi$ is injective.

Surjective: Let $g \in G$ by ii. $g = hk$ for $h \in H$ and $k \in K$ thus $\varphi(h, k) = hk = g$ thus surjective

Thus φ is an isomorphism

- (iii. $\implies$ i.)

Assume iii.

Let $g \in G$ and $k \in K$

By iii. since $H \times K$ is isomorphic to G this implies for all $g \in G$ can be written uniquely as a product hk , with $h \in H$ and $k \in K$

Let $g \in G$ and $h_1 \in H$

gh_1g^{-1} by iii. $g = hk$ for $h \in H$ and $k \in K$

$hkh_1k^{-1}h^{-1}$

Consider $kh_1k^{-1} \in G \implies kh_1k^{-1} = k\varphi(h_1, k^{-1}) = k\varphi((e, k^{-1})(h_1, e)) = k\varphi(e, k^{-1})\varphi(h_1, e) = kk^{-1}h_1 = h_1 \in H$

Thus $kh_1k^{-1} = h_1 \in H \implies gh_1g^{-1} \in H$ thus H is normal in G

Let $g \in G$ and $k_1 \in K$

gk_1g^{-1} by iii. $g = hk$ for $h \in H$ and $k \in K$

$hkk_1k^{-1}h^{-1}$. Let $l = kk_1k^{-1}$

$hllh^{-1} = \varphi(h, l)\varphi(h, e)h^{-1} = lhh^{-1} = l \in K \implies K \trianglelefteq G$

- (b) If G is an internal direct product of H and K then $G \cong H \times K$

Problem 5

- (a) $\varphi : \{\text{subgroups of } G \text{ containing } N\} \rightarrow \{\text{subgroups of } G/N\}$ s.t. $\varphi(H) = \phi(H) = H/N$
 $\varphi^{-1} : \{\text{subgroups of } G/N\} \rightarrow \{\text{subgroups of } G \text{ containing } N\}$ s.t. $\varphi^{-1}(\bar{H}) = \phi^{-1}(\bar{H})$

Let $H \leq G$

$$\varphi^{-1}(\varphi(H)) = \varphi^{-1}(H/N) = H$$

Let $\bar{H} \leq G/N$

$$\varphi(\varphi^{-1}(\bar{H})) = \varphi(\phi(\bar{H})) = \bar{H}$$

Thus φ is bijective since it has an inverse

Let $H \trianglelefteq G$ and $H \supseteq N$ consider $\varphi(H) = H/N$

Let $gN \in G/N$ and $hN \in H/N$

$$gNhNg^{-1}N = ghg^{-1}N \text{ since } H \trianglelefteq G \implies ghg^{-1} \in H \implies gNhNg^{-1}N = ghg^{-1}N \in H/N$$

Let $\bar{H} \trianglelefteq G/N$ consider $\varphi^{-1}(\bar{H})$

Let $g \in G$ and $h \in \varphi^{-1}(\bar{H})$

$$\phi(ghg^{-1}) = ghg^{-1}N \in \bar{H} \text{ since } \bar{H} \text{ is normal} \implies ghg^{-1} \in \varphi^{-1}(\bar{H}) \implies \varphi^{-1}(\bar{H}) \trianglelefteq G$$

- (b) Let $h \in H$ and $n \in N$

$$hnh^{-1} \text{ since } h \in H \subseteq G \text{ and } N \trianglelefteq G \implies hnh^{-1} \in N \implies N \trianglelefteq H$$

By (a) since $H \trianglelefteq G \implies H/N \trianglelefteq G/N$

Let $\varphi : G/N \rightarrow G/H$ s.t. $\varphi(gN) = gH$

Well Defined: Let $xN = yN \implies xy^{-1} \in N \subseteq H$

$$\implies xy^{-1}H = H \implies xH = yH \implies \varphi(xN) = \varphi(yN)$$

Kernel: Let $gN \in \ker(\varphi)$ s.t. $g \in G$

$$\varphi(gN) = gH = H \implies g \in H \implies gN \in H/N$$

Let $hN \in H/N$ s.t. $h \in H$

$$\varphi(hN) = hH = H \implies hN \in \ker(\varphi)$$

Thus $\ker(\varphi) = H/N$

Surjective: Let $gH \in G/H$ s.t. $g \in G$

Consider: $\varphi(gN) = gH$ thus surjective

By the first isomorphism theorem $\frac{G/N}{H/N} \cong G/H$

Problem 6

Let G be order 8.

By Lagrange's theorem G can only have terms of order 1,2,4,8

1. If G has an element $g \in G$ s.t. order of g is 8.

$\langle g \rangle \leq G$ but since $|\langle g \rangle| = |G| \implies \langle g \rangle = G \implies G$ is cyclic thus $G \cong C_8$

2. If G does not have an element of order 8 then G has elements of order 1,2,4 By Cauchy's theorem since $2 \mid 8 = |G| \implies \exists a \in G$ s.t. $\text{ord}(a) = 2$ and let $\langle a \rangle = K \cong C_2$

Consider G/K by Lagrange's theorem $|G/K| = 4$ thus proven in class $G/K \cong C_4$ or $G/K \cong C_2 \times C_2$